

Lesson 8: Conditionals Make

45 minutes

Overview

Using Programming Patterns and a step-by-step approach students make their own version of a Museum Ticket Generator app. At the beginning of the lesson students are able to explore a working version of the app. They are then given the design elements of the app but begin with a blank screen. A progression of levels guides students on the high level steps they should use to develop their app but leaves it to them to decide how to write the code. At the end students submit their apps and answer a free-response question in the style of a Create performance task written response prompt. Student apps and written responses can be assessed using a provided rubric.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

► AP - Algorithms & Programming

Agenda

Warm Up (2 minutes)

Intro the Project

Activity (38 minutes)

Try the Museum Ticket Generator

Make the Museum Ticket Generator App

Wrap Up (5 minutes)

Assessment: Make Project

Objectives

Students will be able to:

- Implement programming patterns with boolean expressions and conditionals statements to develop a functioning app
- Recognize the need for programming patterns with Boolean expressions and conditional statements as part of developing a functioning app
- Use debugging skills as part of developing an app
- Write comments to clearly explain both the purpose and function of different segments of code within an app

Preparation

- Review the different steps students will be asked to complete as they build the app
- Review the information covered in the slides
- Review the Free-Response Supports and consider which, if any, supports are suitable for your classroom.
- Check the "[**Teacher's Lounge**](#)" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our [**Virtual Lesson Modifications**](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **App Code Support - Museum Ticket Generator App** - Handout
[▼ Make a Copy](#)
- **Conditionals Make** - Slides
[▼ Make a Copy](#)
- **Writing Support - Museum Ticket Generator App** - Handout
[▼ Make a Copy](#)

For the students

- **CSP Conditionals Make - Museum Ticket Generator App** - Activity Guide
[▼ Make a Copy](#)

Teaching Guide

Warm Up (2 minutes)

 Teaching Tip 

Short Intro: The Warm Up today is short and light. Students should spend the maximum amount of time working on their projects.

Intro the Project

Remarks

For the past few days, we've learned a lot about using conditional statements to help apps make decisions. In today's Make Project you'll be practicing both making flow charts and writing complex conditionals statements as you build a Museum Ticket Generator app.

Activity (38 minutes)

Try the Museum Ticket Generator

Group: Make a determination as to whether this project will be completed in pairs or individually. You may even choose to let students decide.

Distribute: Make sure students have access to scrap paper and pencils / paper for drawing flow charts.

 **Level 1 - Explore:** Have students explore the working Museum Ticket Generator App in Level 1. If students are not working in pairs they should still discuss with a neighbor.

 Teaching Tip 

This lesson is an opportunity for students to take on the "blank screen" and build the code that runs an app entirely from scratch. Guidance provided throughout the lesson helps students break down the large task of "building an app" into more incremental steps that they can use on future projects, including this unit's final project and the Create PT.

Discuss:

- *What does this app do?*
 - Generates a ticket to a museum based on input data from the user.
- *What are the inputs?*
 - The dropdowns for age and day of the week
 - The discount code text field
 - The button to generate the ticket
- *What are the outputs?*
 - The ticket generated at the bottom of the screen
- *What variables do you think would be necessary for this app to work?*
 - The input information listed above will all need variables (age, day, discountCode), but they may want variables for other information calculated during the app.
- *What kinds of conditional logic do you think are necessary to make it work?*
 - The pricing rules help you know what to write. They'll definitely need to use if-else-if statements and AND / OR operators in order to implement these rules since they include multiple conditions to check.



Try to the Museum Ticket Generator App

Make the Museum Ticket Generator App

Teaching Tip

Supporting Students: While students are working on their apps, circulate the room and check in with students who need a little help. Encourage students to collaborate and discuss bugs with each other.

Debugging: Review with students steps they can use to debug if they get stuck:

- Run the code on turtle mode
- Add the variables to the watcher
- Explain the code to a friend

Remarks

Now let's build the this app. The screen has been set up for you - it's your job to add the code!

Distribute: Give students copies of **Conditionals Make Activity Guide** if you will be using it during the class.

Do This: Direct students to level two where they complete the Museum Ticket Generator App. Based on the needs of your classroom decide whether you will collectively go through the activity guide or have students complete it individually. Afterwards give them time to work on their projects and circulate the room to offer support. Students who finish early can work on the extensions suggested in the activity guide.

Submit: Encourage students to check the rubric on the last page of the Activity Guide before submitting.

Teaching Tip

Adapting Make Lessons - You may decide that you'd like to have your students write the code for a modified or entirely different app using the same skills. This might help you differentiate your instruction or provide you with a way to reassess students. Check out the [r customizemakelessons_how-to-guide/csp/2025] for guidance on how to get started with this.

Wrap Up (5 minutes)

Assessment Opportunity

Create Performance Task Written Response Practice

In order to help prepare students for the new component of the AP Create Performance Task, there will be several opportunities throughout the curriculum to practice creating written responses aligned with the learning objectives that might be assessed on the exam.

Sample Prompt: *Examine one procedure (function) that is used in your program. If there's an algorithm that employs relational operators, explain its functionality. If not, how could the inclusion of relational operators optimize or modify the outcome?*

Unit Journal Template

Remarks

Great work on your Museum Ticket Generator app today. Before wrapping up, we will be practicing writing for the style of questions that are asked as part of the Create performance task on the AP exam. Respond to the question in Step 5 in your activity guide, and submit it along with your project.

Do This: Complete the written response in Step 5 of the Activity Guide.

Teaching Tip

Each student, in every classroom, will come to the written response with a different level of comfort, ability, and familiarity with writing for these style of questions. As such, consider offering one or more of the following supports:

Writing Support: Provide students with these sentence starters for guiding their response.

App Code Support: For students who did not fully complete their app, provide them with this snippet from the exemplar code.

Assessment Opportunity

Student responses can be used as a formative assessment to evaluate their understanding of connections between user interface features and functionality. Review their explanations for clarity, accuracy, and alignment with the AP exam's expectations. The provided rubric can be used to guide this review, and to help provide

students with feedback to guide their learning and improvement

Assessment: Make Project

Use the rubric provided with the project to assess student projects.



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